

$(G, \cdot)$ Group. $\cdot: G \times G \rightarrow G$ satisfying $\begin{cases} \text{associative} \\ \text{inverse} \\ \text{identity.} \end{cases}$

$\ast: G \times A \rightarrow A$ Group Action.

$N \leq G$ is Normal

$$\Leftrightarrow \forall g \in G, n \in N, gng^{-1} \in N.$$

$$\Leftrightarrow gN = Ng$$

Note: Since $\sigma_g^{-1} = (\sigma_g)^{-1}$, $\sigma_g \in S_A$.

$$\begin{cases} \sigma_g(a) = g \cdot a, \\ (g_1 g_2) \cdot a = g_1 \cdot (g_2 \cdot a) \end{cases} \quad \forall g_1, g_2 \in G, a \in A$$

$$\begin{cases} \sigma_g: A \rightarrow A: a \mapsto g \cdot a \\ \varphi: G \rightarrow S_A: g \mapsto \sigma_g \end{cases} \Rightarrow g \cdot a = \varphi(g)(a)$$

stabilizer of a in G : $G_a := \{g \in G \mid g \cdot a = a\}$

Note: $G_a \leq G$

$\ker \ast \trianglelefteq G$.

kernel of action: $\ker \ast := \{g \in G \mid g \cdot a = a, \forall a \in A\} = \bigcap_{a \in A} G_a$.

Note: $\ker \ast = \ker \varphi$

$\ker \varphi = \{g \in G \mid \varphi(g) = I\}$ (I is identity map on A)

$$= \{g \in G \mid \sigma_g = I\}$$

$$= \{g \in G \mid \forall a \in A, g \cdot a = a\}.$$

$$= \ker \ast$$

Let $a \sim b \Leftrightarrow \exists g \in G$ s.t. $a = g \cdot b$. Then $\sim$ is equivalent relation, and

$C_a = [a]$ satisfies:

$|C_a| = |G|/|G_a|$, because $\varphi_a: C_a \rightarrow \{g G_a \mid g \in G\}: g \cdot a \mapsto g G_a$ is bijective.

Thm. Let G be a Group, $H \leq G$ be a subgroup. Put $A = \{gH \mid g \in G\}$.

$\ast: G \times A \rightarrow A$ be a G -Action.

$\Pi_H: G \rightarrow S_A: g \mapsto \sigma_g$ where $\sigma_g: A \rightarrow A: xH \mapsto g \cdot xH$.

1. $\forall a \in A, C_a = A$. That is, this Action is transitive.

$$2. G_{1H} = \{g \in G \mid g \cdot (1 \cdot H) = 1 \cdot H\} = H.$$

3. $\ker \Pi_H = \bigcap_{x \in G} xHx^{-1}$ is the largest Normal subgroup G contained in H .

Let G be a group. $A \subseteq G$ subset.

$$N_G(A) = \{g \in G \mid gAg^{-1} = A\},$$

$$C_G(A) = \{g \in G \mid ga = ag, \forall a \in A\}.$$

$$C_G(A) \leq N_G(A) \leq G$$

$S \leq G \Rightarrow N_G(S)$ is the largest subgroup s.t. $S \trianglelefteq N_G(S)$.

$S, T \leq G$ are conjugate in G if: $\exists g \in G$ s.t. $T = gSg^{-1}$.

Define an act $\ast: G \times \mathcal{P}(G) \rightarrow \mathcal{P}(G): (g, S) \mapsto gSg^{-1}$.

Then, $G_S = N_G(S)$.

Moreover, An orbit of S , $\mathcal{O}_S = \{gS \mid g \in G\}$ has size as $|G:N_G(S)|$.

$$(|\mathcal{O}_S| = |G:G_S|)$$

$g \cdot a \mapsto gAg^{-1}$ is bijection.

Note: $G = C_G(z_i) \Leftrightarrow z_i \in Z(G)$.

That is, $z_i \notin Z(G) \Rightarrow |C_G(z_i)| < |G|$.

Class Equation.

$$|G| = |Z(G)| + \sum_{z_i \text{ is}} |G:C_G(z_i)|$$

representative of distinct conjugacy classes of G ,

$$\text{Aut}(G) \stackrel{\text{def}}{=} \{ \varphi: G \rightarrow G \mid \varphi \text{ is isomorphism} \}$$

[Note: $\sigma_g: H \rightarrow H; h \mapsto ghg^{-1}$ is Automorphism]

$$ghg^{-1} = gh_2g^{-1} \Rightarrow h_1 = h_2$$

For $h \in H$, $g^{-1}hg \in H$. Thus,

$$\sigma_g(g^{-1}hg) = g(g^{-1}hg)g^{-1} = h.$$

$$\text{Finally, } \sigma_g(h_1h_2) = gh_1h_2g^{-1} = gh_1g^{-1}gh_2g^{-1} = \sigma_g(h_1)\sigma_g(h_2)$$

Let $H \leq G$. Define a Action of G on H :

$$G \times H \rightarrow H; (g, h) \mapsto ghg^{-1}.$$

Note This Action is well-defined by Normality.

Now the permutation representation afforded by this Action

$$\varphi: G \rightarrow S_H: g \mapsto \sigma_g \quad \text{where } \sigma_g: H \rightarrow H; h \mapsto ghg^{-1}.$$

while $\varphi[G] = \text{Aut}(H) \leq S_H$. And $\ker \varphi = C_G(H)$. [Thus $G/\ker \varphi = G/C_G(H) \cong \text{Aut}(H)$]

Proof. Enough to show that $\ker \varphi = C_G(H)$.

$$\begin{aligned} G/\ker \varphi &\cong \varphi[G] \\ &\cong G/C_G(H) \cong \text{Aut}(H) \end{aligned}$$

$$\ker \varphi := \{ g \in G \mid \sigma_g = I \}$$

$$= \{ g \in G \mid \forall h \in H, ghg^{-1} = h \} = C_G(H).$$

[Corollary. Suppose $g \in G$, $K \leq G$. Then $K \cong gKg^{-1}$.

Proof. $K \rightarrow gKg^{-1}; x \mapsto gxg^{-1}$ is isomorphism.

[Corollary. Let G be a group, $H \leq G$. Then,

$$N_G(H)/C_G(H) \leq \text{Aut}(H).$$

In particular, $G/Z(G) \leq \text{Aut}(G)$.

$$(C_G(H) < N_G(H))$$

Proof. Since $H \trianglelefteq N_G(H) \leq G$, and $C_{N_G(H)}(H) = C_G(H) \cap N_G(H) = C_G(H)$,

$$N_G(H)/C_{N_G(H)}(H) = N_G(H)/C_G(H) \cong \varphi[N_G(H)] \leq \varphi[G] = \text{Aut}(H).$$

[Def. Let G be a group. For $g \in G$, $\varphi_g: G \rightarrow G; a \mapsto ga g^{-1}$ is Automorphism.

This φ_g is called inner Automorphism denote set of inn. Aut

$$\text{Inn}(G) \leq \text{Aut}(G).$$

Def. A subgroup $H \leq G$ is called characteristic in G if:

For any $\sigma \in \text{Aut}(G)$, $\sigma[H] = H$.

Denote $H \text{ char } G$.

Thm. Let $H \leq G$. Then,

1). $H \text{ char } G \Rightarrow H \trianglelefteq G$.

2). If H is the unique subgroup of G of given order, then $H \text{ char } G$.

3). $K \text{ char } H$ and $H \trianglelefteq G$, then $K \trianglelefteq G$.

Proof. Suppose $H \text{ char } G$. Let $g \in G$, $h \in H$. Then, for inner-Auto ϕ_g ,
 $ghg^{-1} = \phi_g(h) \in H$, by definition.

Suppose $K \leq G$ and $|K| = |H| \Rightarrow K = H$.

Then, for any $\sigma \in \text{Aut}(G)$, $|\sigma[H]| = |H|$, thus $\sigma[H] = H$.

Suppose $K \text{ char } H$ and $H \trianglelefteq G$.

Let $g \in G$. Then, $\phi_g: H \rightarrow H; h \mapsto ghg^{-1}$
is well-defined Automorphism of H .

Thus, $\phi_g[K] = gKg^{-1} = K$, thus $K \trianglelefteq G$.

Thm. $\text{Aut}(\mathbb{Z}_n) \cong \mathbb{Z}_n^\times$, which is an Abelian Group of $\varphi(n)$.

Def. Let G be a Group, $x, y \in G$, $A, B \subseteq G$.

1) $[x, y] \stackrel{\text{def}}{=} x^{-1}y^{-1}xy$, commutator of x and y .

2) $[A, B] \stackrel{\text{def}}{=} \langle [a, b] \mid a \in A, b \in B \rangle$

3) $G' = \langle [x, y] \mid x, y \in G \rangle$.

Prop. Let G be a Group, $x, y \in G$, $H \leq G$.

1) $xy = yx[x, y]$. In particular, x, y are commute iff $[x, y] = 1$.

2) $H \trianglelefteq G$ iff $[H, G] \leq H$.

3) $\forall \sigma \in \text{Aut}(G)$, $\sigma[x, y] = [\sigma(x), \sigma(y)]$

4) G' char G

5) G/G' is the largest Abelian Quotient of G , that is,

$H \trianglelefteq G$ and G/H Abelian iff $G' \leq H$.

6. If $\varphi: G \rightarrow A$ is Homomorphism into an Abelian Group A , then $G' \leq \ker \varphi$.

Proof. 1) is trivial, let's show the 2).

$H \trianglelefteq G \Leftrightarrow \forall g \in G, hg^{-1}h^{-1} \in H$.

$\Leftrightarrow \forall g \in G, hg^{-1}h^{-1} = ghg^{-1}h^{-1} \in H$

$\Leftrightarrow \forall g \in G, h \in H, [h, g] \in H$

$\Leftrightarrow [H, G] \leq H$.

3). $\sigma[x, y] = \sigma(x^{-1}y^{-1}xy) = \sigma(x)^{-1}\sigma(y)^{-1}\sigma(x)\sigma(y) = [\sigma(x), \sigma(y)]$

4). Directly, since 3).

5) First, show Normality. By 2), $G' \trianglelefteq G$ iff $[G', G] \leq G'$. The right term is trivial:

$x \in [G', G] \Leftrightarrow x = [x_1, y_1] \cdots [x_n, y_n]$ for some $x_i \in G'$, $y_i \in G$.

Clearly $x_i \in G$, $x \in G'$.

Second, G/G' is Abelian, because: let $g_1G', g_2G' \in G/G'$ be given.

Then, $g_1G' \cdot g_2G' = g_1g_2G' = g_2g_1[x_1, x_2]G' = g_2g_1G' = g_2G' \cdot g_1G'$.

Finally, to show the largest: let $H \trianglelefteq G$ and G/H Abelian. Then, for any $x, y \in G$,

$1 \cdot 1 = (x^{-1}H)(y^{-1}H)(xH)(yH) = [x, y]H \Leftrightarrow [x, y] \in H$. Thus $G' \leq H$.

Conversely, suppose $G' \leq H$. Since G/G' is Abelian, Every subgroup of G/G' is Normal.

That is, $G/H \trianglelefteq G/G'$, this implies $H \trianglelefteq G$.

Recall: for $H, K \leq G$, if each are finite,

$$|HK| = \frac{|H||K|}{|H \cap K|}$$

Note: $| \cdot | \rightarrow g | \cdot | : H \rightarrow gHg^{-1}$ bijective.
 $a \sim b \Leftrightarrow ab^{-1} \in H$ is Equi. rel.

proof. $HK = \bigcup_{h \in H} hK$ clearly. And, for $h_1, h_2 \in H$,

$$h_1K = h_2K \Leftrightarrow h_1h_2^{-1} \in H \cap K \Leftrightarrow h_1(H \cap K) = h_2(H \cap K)$$

$$\text{Now, } |HK| = \left| \bigcup_{h \in H} hK \right| = |H : H \cap K| \cdot |K| = \frac{|H|}{|H \cap K|} \cdot |K|$$

Thm. Let $K, H \leq G$, $x \in HK$.

$$|\{(h, k) \mid h \in H, k \in K, hk = x\}| = |H \cap K|. \quad |HK| = \frac{|H||K|}{|H \cap K|}$$

proof. Let $S := HxK$. Then, clearly $|S| = |H||K|$.

Define a relation on S as:

$$(h_1, k_1) \sim (h_2, k_2) \Leftrightarrow h_1k_1 = h_2k_2$$

Then, clearly $\sim$ is Equi. rel. on S . Now, consider a Partition:

$$S/\sim = \{[h, k] \mid h \in H, k \in K\}$$

And, for any classes $[h, k]$, there exists a bijection from $[(1, 1)]$ into $[h, k]$:

$$\varphi: [(1, 1)] \rightarrow [h, k] : (a, a^{-1}) \mapsto (ha, a^{-1}k)$$

$$\text{well-defined: } [(1, 1)] = \{(h, k) \mid h \in H, k \in K, hk = 1\}$$

$$= \{(a, a^{-1}) \mid a \in H \cap K\}. \text{ And, } ha a^{-1}k = hk$$

$$\text{injective: } (ha, a^{-1}k) = (hb, b^{-1}k) \Rightarrow a = b$$

surjective: Let $(h_1, k_1) \in HxK$ be a tuple s.t. $h_1k_1 = x$.

Then, $h^{-1}h_1 = k_1k^{-1} \in H \cap K$. Now,

$$\varphi(h^{-1}h_1, (h^{-1}h_1)^{-1}) = \varphi(h^{-1}h_1, k_1k^{-1}) = (hh^{-1}h_1, k_1k^{-1}k) = (h_1, k_1)$$

Consequently, for any class $[h, k] \in S/\sim$,

$$|S| = |H| \cdot |K| = |S/\sim| \cdot |[h, k]| = |HK| \cdot |[h, k]| = \frac{|H||K|}{|H \cap K|} \cdot |[h, k]|$$

$$\Rightarrow |[h, k]| = |H \cap K|$$

Recognition Theorem.

Suppose $H, K \leq G$ are Normal Subgroup s.t. $H \cap K = \{1\}$.

Then, $HK \cong H \times K$.

Proof. $\varphi: HK \rightarrow H \times K: h_1 k_1 \mapsto (h_1, k_1)$ is isomorphism, because:

1) $HK \leq G$.

$$HK = \bigcup_{h \in H} hK = \bigcup_{h \in H} K h = KH$$

Now, for any $h_1 k_1, h_2 k_2 \in HK$, $\underbrace{h_1 k_1 k_2^{-1} h_2^{-1}}_{\in KH = HK} \in HK$.

2). $\forall h \in H, k \in K, hk = kh$.

Let $h \in H, k \in K$. Since Normality, $\begin{cases} k^{-1} h k \in H \Rightarrow h^{-1} k^{-1} h k \in H \\ h^{-1} k^{-1} h \in K \Rightarrow h^{-1} k^{-1} h k \in K \end{cases}$

That is, $[h, k] \in H \cap K = \{1\}$, h, k are commute.

3). Well-defined of HK .

By Theorem, each element of HK has the unique representation form

4). Injective. and surjective.

Trivial.

5) Homomorphism.

$$\varphi(h_1 k_1 \cdot h_2 k_2) = \varphi(h_1 h_2 k_1 k_2) = (h_1 h_2, k_1 k_2) = (h_1, k_1) \cdot (h_2, k_2) = \varphi(h_1 k_1) \cdot \varphi(h_2 k_2).$$

Def. If G is a Group, and $H, K \trianglelefteq G$ are Normal Subgroup with $H \cap K = \{1\}$,

$H \times K$ is called internal direct product, of H and K .

$H \times K$ is called external direct product.

Def. Let G be a Group.

A Chain of subsets $\{A_i\}_{i=0}^n$ is called Subgroup series if:

$$1 = A_0 \trianglelefteq A_1 \trianglelefteq \dots \trianglelefteq A_{n-1} \trianglelefteq A_n = G.$$

A s.g. series is called Subnormal series if:

$$1 = A_0 \trianglelefteq A_1 \trianglelefteq \dots \trianglelefteq A_{n-1} \trianglelefteq A_n = G.$$

And A_{i+1}/A_i ($0 \leq i \leq n-1$) is called factor group.

Especially, if $A_i \trianglelefteq G \quad \forall 0 \leq i \leq n$, then it is called Normal Series.

A Subnormal Series $\{N_i\}_{i=0}^n$ is called Composition series if

for all $0 \leq i \leq n-1$, N_{i+1}/N_i is a Simple.

A Group G is called Solvable if, there is a Subnormal series $\{H_i\}_{i=0}^n$ s.t.
for all $0 \leq i \leq n-1$, H_{i+1}/H_i is a Abelian.

⌈ p -Group.

Thm. Let G be a Group of order p^n for some prime p , $n \in \mathbb{N}$.

Then, G has a Normal Series $\{P_i\}_{i=0}^n$ such that

$$1 = P_0 \triangleleft P_1 \triangleleft \dots \triangleleft P_{n-1} \triangleleft P_n = G. \quad (|P_i| = p^i)$$

Proof. First, $Z(G) \neq 1$ because The Class Equation: if $Z(G) = 1$,

$$p^n = |G| = |Z(G)| + \sum |G : G(g_i)| = 1 + \sum p^{r_i} \quad (1 \leq r_i \leq n-1)$$

This implies p divides 1, Contradiction.

Now, Using induction:

For $n=2$,

$|G/Z(G)| = p$ or 1, thus it must be cyclic, this implies G is Abelian.

By Cauchy's Thm, G has a subgroup of order p , thus Claim is satisfied.

For $n \geq 2$, suppose the claim is true.

Let $|G| = p^{n+1}$. Since $1 \subsetneq Z(G)$ and Cauchy Thm,

there exists a Normal Subgroup $N \trianglelefteq G$ in $Z(G)$ with order p .

Now, G/N is a Group of order p^n , by hypothesis, G/N has a Normal Series:

$$1 = N/N \triangleleft N_1/N \triangleleft \dots \triangleleft N_{n-1}/N \triangleleft N_n/N = G/N.$$

The fourth Isomorphism gives $N_{n-1} \trianglelefteq G$ and $|N_{n-1}| = p^n$, the claim is proved.

Lemma. $G/Z(G)$ is cyclic $\Leftrightarrow G$ is Abelian. D

Proof. Suppose $G/Z(G)$ is cyclic. Put $xZ(G)$ be a generator of $G/Z(G)$.

Let $a, b \in G$ be given. Then, for some $m, n \in \mathbb{Z}$,

$$aZ(G) = x^m Z(G), \quad bZ(G) = x^n Z(G).$$

Now, $ax^m, bx^n \in Z(G)$.

$$ab = ax^{-m} x^m bx^{-n} x^n = x^{m+n}. \quad bx^{-n} ax^{-m} = ba.$$

Prop. Dummit, #3.4.1.

If G is Abelian Simple Group, then $G \cong \mathbb{Z}_p$ for some prime p . (Also converse is true)

Proof. Since G is Abelian, every proper non-trivial subgroup $1 \subsetneq H \subsetneq G$ is Normal.

Simplicity of G gives G has no proper non-trivial subgroup.

That is, for any nonzero $g \in G$, $\langle g \rangle = G$.

Now, $G \cong \mathbb{Z}_k$, for some $k \in \mathbb{N}$ or ∞ .

If $G \cong \mathbb{Z}$, then it has $2\mathbb{Z}$ as subgroup, thus $G \cong \mathbb{Z}_n$ for some $n \in \mathbb{N}$.

If n is not prime, it has $m\mathbb{Z}$ as subgroup, thus $G \cong \mathbb{Z}_p$ for some prime p .

Lemma. Let G be a group, $N \trianglelefteq G$ be a Normal Subgroup. Then,

G/N is Simple $\iff N$ is Maximal Normal subgroup of G .

Proof. By 4th isomorphism Theorem,

G/N is Simple $\Leftrightarrow G/N$ has no proper non-trivial Normal Subgroup

$\Leftrightarrow$ There is no Normal $\mathbb{Z} \triangleleft N \triangleleft H \triangleleft G$.

$\Leftrightarrow N$ is Maximal Normal.

First Jordan-Hölder Theorem.

Non-trivial finite Group has a Composition Series.

Lemma. Every Non-trivial Finite Group has a Maximal Normal Subgroup.
proof. if G is simple, then $1 \triangleleft G$ is Maximal Normal.

Suppose G is not simple. Then, G has at least one Non-trivial proper Normal N .

If N is Maximal Normal, there is nothing to prove.

If N is not Maximal Normal, there is proper Normal Super set
 $N < N_1 \triangleleft G$.

This process must terminate in finite, being G is finite,
 G has a Maximal Normal Subgroup.

Proof of Jordan-Hölder

Prop. If G is finite Group, then

G is solvable iff G has Composition series s.t. all factors are Abelian.
Proof. $\Leftarrow$ is trivial.

Suppose that G is Solvable. That is, G has a subnormal series $\{H_i\}_{i=0}^n$ s.t.

$$1 = H_0 \triangleleft H_1 \triangleleft \dots \triangleleft H_{n-1} \triangleleft H_n = G, \quad H_{i+1}/H_i \text{ is Abelian.}$$

For each H_{i+1}/H_i ($0 \leq i \leq n-1$), H_{i+1}/H_i has a Composition series $\{H_{i,j}/H_i\}_{j=0}^{r_i}$ s.t.

$$1 = H_i/H_i = H_{i,0}/H_i \triangleleft H_{i,1}/H_i \triangleleft \dots \triangleleft H_{i,r_i-1}/H_i \triangleleft H_{i,r_i}/H_i = H_{i+1}/H_i.$$

by Jordan-Hölder Thm. That is,

$$(H_{i,r_i+1}/H_i)/(H_{i,0}/H_i)$$

is Simple and Abelian. The 4th Isomorphism Thm gives a series $\{H_{i,j}\}_{j=0}^{r_i}$ s.t.

$$H_i = H_{i,0} \triangleleft H_{i,1} \triangleleft \dots \triangleleft H_{i,r_i-1} \triangleleft H_{i,r_i} = H_{i+1}$$

Indeed, all factor of this is Simple and Abelian, because the 3rd Isomorphism Thm gives:

$$(H_{i,r_i+1}/H_i)/(H_{i,0}/H_i) \cong H_{i,r_i+1}/H_{i,0}$$

Consequently,

$$1 = H_0 = H_{0,0} \triangleleft H_{0,1} \triangleleft \dots \triangleleft H_{0,r_0} = H_{1,0} \triangleleft H_{1,1} \triangleleft \dots \triangleleft H_{1,r_1} = H_2 = G.$$

is a Composition series satisfying every factor is Abelian.

(Actually, Simple Abelian group is isomorphic to $\mathbb{Z}_p$, every factor is cyclic).

Isomorphism Theorem.

1st. $G/\ker \phi \cong \phi[G]$ for any Homomorphism $\phi: G \rightarrow G'$,

2nd. If $A, B \leq G$ and $A \leq N_G(B)$, then $AB \leq G$ and $B \trianglelefteq AB$, $A \cap B \trianglelefteq A$, and

$$AB/B \cong A/A \cap B.$$

Proof. $AB = \bigcup_{a \in A} aB = \bigcup_{a \in A} Ba = BA$. Thus $AB \leq G$.

And, $A, B \leq N_G(B)$ implies $AB \leq N_G(B)$, thus $B \trianglelefteq AB$.

$A \cap B \trianglelefteq A$ is clear. Now, a Map

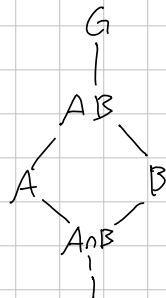
$$\phi: A \rightarrow AB/B: a \mapsto aB$$

is onto Homomorphism with $\ker \phi = A \cap B$.

3rd. If $H, K \leq G$ and $H \leq K$, then $K/H \leq G/H$ and

$$[G/H]/[K/H] \cong G/K.$$

Proof. Take $\phi: G/H \rightarrow G/K: gH \mapsto gK$ as onto Homomorphism with $\ker \phi = K/H$.



4th. Let G be a Group, $N \trianglelefteq G$ be a Normal subgroup.

Denote $\mathcal{S} := \{A \leq G \mid N \leq A \leq G\}$ and $\overline{\mathcal{S}} := \{\overline{A} \leq G/N \mid 1 = N/N \leq \overline{A} \leq G/N\}$.

Then, a Map $\mathcal{Q}: \mathcal{S} \rightarrow \overline{\mathcal{S}}: A \mapsto A/N$ is bijection satisfying: for any $N \leq A, B \leq G$,

$$1) A \leq B \iff \mathcal{Q}(A) \leq \mathcal{Q}(B)$$

$$2) \text{ If } A \leq B, \text{ then } |B:A| = |\mathcal{Q}(B):\mathcal{Q}(A)|.$$

$$3) \mathcal{Q}(\langle A, B \rangle) = \langle \mathcal{Q}(A), \mathcal{Q}(B) \rangle$$

$$4) \mathcal{Q}(A \cap B) = \mathcal{Q}(A) \cap \mathcal{Q}(B)$$

$$5) A \trianglelefteq B \iff \mathcal{Q}(A) \trianglelefteq \mathcal{Q}(B)$$

Proof. Injective: Suppose $\mathcal{Q}(A) = \mathcal{Q}(B) \Leftrightarrow A/N = B/N$.

Let $a \in A$. By assumption, $aN = bN$ for some $b \in B$. Thus, $ab^{-1} \in N$.

Since $N \leq B$, $ab^{-1} \cdot b \in B$. That is, $A \leq B$. Similarly, $A \geq B$. Thus, $A = B$.

Surjective: Let $\overline{A} \in \overline{\mathcal{S}}$ be given. Consider a Map $\pi: G \rightarrow G/N, g \mapsto gN$.

Since $\overline{A} \leq G/N$, the Preimage $\pi^{-1}[\overline{A}]$ is also subgroup of G containing N .

$$\text{Now, } \mathcal{Q}(\pi^{-1}[\overline{A}]) = \pi^{-1}[\overline{A}]/N = \pi^{-1}[\pi^{-1}[\overline{A}]]_{\uparrow N} = \overline{A}$$

Proof 1) $\Rightarrow$ is trivial. Suppose $A/N \leq B/N$. Then, for any $x \in A$, $xN \in B/N$.

That is, $x = bn$ for some $b \in B, n \in N$. Since $N \leq B$, $x = bn \in B$.

Proof 5). Let $A \leq B$. For any $xN \in A/N$ and $bN \in B/N$, $bx b^{-1}N \in A/N$.

Let $A/N \trianglelefteq B/N$. For any $x \in A, b \in B$, $bx b^{-1}N \in A/N$, thus $bx b^{-1} \in A$.

Proof 2). Let

$$\psi: \{gA \mid g \in B\} \rightarrow \{gN\mathcal{Q}(A) \mid gN \in \mathcal{Q}(B)\}: gA \mapsto gN\mathcal{Q}(A).$$

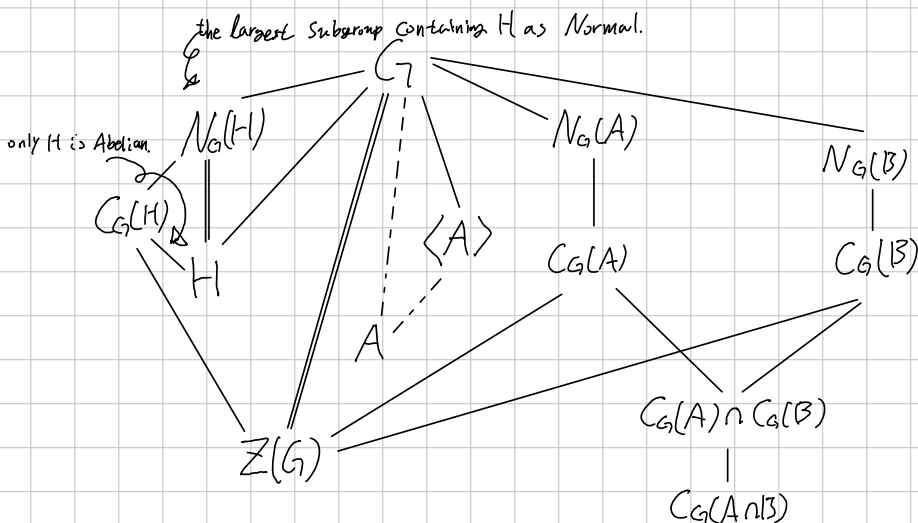
Then, ψ is well-defined injection:

$$g_1N\mathcal{Q}(A) = g_2N\mathcal{Q}(A) \Leftrightarrow g_1g_2^{-1}N \in \mathcal{Q}(A) = A/N \Leftrightarrow g_1g_2^{-1} \in A \Leftrightarrow g_1A = g_2A$$

And surjection is clear.

— Subgroup
 == Normal
 --- Subset

*: $G \times A \rightarrow A$ is Group Action.



$$C_G(A \cap B) \leq C_G(A) \cap C_G(B)$$

$$N_G(A \cap B)$$

Let $x \in \underline{\quad}$. Then, $x(A \cap B)x^{-1} = A \cap B$

Let $x \in N_G(A \cap B)$ given

then, $\forall h \in A \cap B, \exists y x^{-1} \in A \cap B,$

Thm. Every finitely generated group has a Maximal Subgroup.

Proof. Let G be a group s.t.

$$G = \langle x_1, \dots, x_n \rangle.$$

Let $H = \langle x_1, \dots, x_{n-1} \rangle$. Then, H is a Maximal,

Butterfly Lemma.

Let G be a Group, $H, K \leq G$, $H^* \trianglelefteq H$ and $K^* \trianglelefteq K$. Then, the following statements hold:

$$1) H^*(H \cap K^*) \trianglelefteq H^*(H \cap K)$$

$$2) K^*(H^* \cap K) \trianglelefteq K^*(H \cap K)$$

$$3) H^*(H \cap K) / H^*(H \cap K^*) \cong K^*(H \cap K) / K^*(H^* \cap K) \cong [H \cap K] / [(H^* \cap K)(H \cap K^*)]$$

Proof. Since $H^* \trianglelefteq H$, for any $h \in H \cap K^*$, $hH^*h^{-1} = H^*$.

[Note: for $A, B, C \leq G$,

$$H^*(H \cap K^*) = \bigcup_{h \in H \cap K^*} H^*h = \bigcup_{h \in H \cap K^*} hH^* = (H \cap K^*)H^*. \text{ Thus } H^*(H \cap K^*) \leq H^*(H \cap K). \quad \varphi(A \cap B) = \varphi A \cap \varphi B,$$

$f: G \rightarrow G: x \mapsto gx$ is injective,

Meanwhile, to show $H^* \cap K \trianglelefteq H \cap K$, let $x \in H \cap K$. Then,

$$x(H \cap K^*)x^{-1} = (xH \cap xK^*)x^{-1} = (H \cap K^*)x^{-1} = Hx^{-1} \cap K^* = H \cap K^*.$$

Similarly $H \cap K^* \trianglelefteq H \cap K$. Since inner product of two normal is normal,

$$L = (H \cap K^*)(H^* \cap K) \trianglelefteq H \cap K.$$

Define a Map:

$$\varphi: H^*(H \cap K) \rightarrow (H \cap K) / L : hx \mapsto xL \quad (h \in H^*, x \in H \cap K).$$

Well-defined: for $h_1, h_2 \in H^*$, $x_1, x_2 \in H \cap K$,

$$h_1x_1 = h_2x_2 \Rightarrow h_2^{-1}h_1 = x_2x_1^{-1} \in H^* \cap (H \cap K) = H^* \cap K \leq L$$

$$\text{Thus } x_2x_1^{-1} \in L \Leftrightarrow x_1L = x_2L.$$

Homomorphism: for $h_1, h_2 \in H^*$, $x_1, x_2 \in H \cap K$,

$$\begin{aligned} \varphi(h_1x_1, h_2x_2) &= \varphi(h_1h_3x_1x_2) \text{ for some } h_3 \in H^*, \text{ since } H^* \trianglelefteq H \\ &= x_1x_2 = \varphi(h_1x_1) \cdot \varphi(h_2x_2). \end{aligned}$$

Onto is trivial. And,

$$\ker \varphi = \{hx \mid h \in H^*, x \in H \cap K, x \in L\} = \{hx \mid h \in H^*, x \in L\} = H^*L = H^*(H \cap K^*).$$

Now, 1st Isomorphism gives us the result.

Schrier Theorem.

Two subnormal series of G have isomorphic refinement.

Proof. Suppose G is a group, and $\{H_i\}_{i=0}^n, \{K_j\}_{j=0}^m$ are subnormal series for G .

For each $0 \leq i \leq n-1$, define $H_{i,j} \stackrel{\text{def}}{=} H_i(H_{i+1} \cap K_j)$ ($0 \leq j \leq m$).

Then, $H_{i,j} \trianglelefteq H_{i,j+1}$ because

$$H_{i,j} = H_i(H_{i+1} \cap K_j) \trianglelefteq H_i(H_{i+1} \cap K_{j+1}) = H_{i,j+1}$$

Now, we obtain

$$1 = H_0 = H_{0,0} \trianglelefteq H_{0,1} \trianglelefteq \dots \trianglelefteq H_{0,m-1} \trianglelefteq H_{0,m} = H_0(H_1 \cap G)$$

$$= H_1 = H_{1,0} \trianglelefteq H_{1,1} \trianglelefteq \dots \trianglelefteq H_{1,m-1} \trianglelefteq H_{1,m}$$

$$= H_2 = \dots$$

$\vdots$

$$= H_{n-1} = H_{n-1,0} \trianglelefteq H_{n-1,1} \trianglelefteq \dots \trianglelefteq H_{n-1,m-1} \trianglelefteq H_{n-1,m}$$

$$= G.$$

In a same sense, define $K_{i,j} \stackrel{\text{def}}{=} K_j(K_{j+1} \cap H_i)$ for $0 \leq j \leq m-1, 0 \leq i \leq n$.

Then, $K_{i,j} \trianglelefteq K_{i,j+1}$. Moreover, the Butterfly Lemma gives:

$$H_{i,j+1}/H_{i,j} = (H_i(H_{i+1} \cap K_{j+1})) / (H_i(H_{i+1} \cap K_j))$$

$$\cong (K_j(K_{j+1} \cap H_{i+1})) / (K_j(K_{j+1} \cap H_i))$$

$$= K_{j+1,i} / K_{j,i}$$

$$Z(G). \quad \varphi_i: G \rightarrow G/Z(G); g \mapsto gZ(G).$$

$$Z_1(G) = \varphi_1^{-1}[Z(G/Z(G))]$$

$$Z_2(G) = \varphi_1^{-1}[Z(G/Z_1(G))]$$

⋮

$$Z_k(G) = \varphi_k^{-1}[Z(G/Z_k(G))]$$

$$\varphi_i: G \rightarrow G/Z_i(G)$$

$$: g \mapsto gZ_i(G)$$

$$Z_i(G) \leq Z_{i+1}(G) \text{ because}$$

$$\text{Let } x \in Z_i(G). \text{ Then, } \varphi_i(x) = xZ_i(G) = 1 \cdot Z_i(G) \in Z(G/Z_i(G))$$

$$D_{2n} = \langle r, s \mid r^n = s^2 = (rs)^2 = 1 \rangle$$

$$S_3 \quad Z(S_3)$$

$$\underline{rs = sr^{-1}}.$$

$$= \{1, (12), (23), (13), (123), (132)\},$$

$$G/Z(G) \text{ cyclic} \Rightarrow G \text{ Abelian.}$$

$$G \text{ not Abelian} \Rightarrow G/Z(G) \text{ not cyclic.}$$

$$|S_4| = 24.$$

$$Z(G) = \bigcap_{g \in G} C_G(g) \quad ?$$

$$rsrs = 1$$

$$srs = r^{-1} = r^3$$

$$D_4$$

$$D_{10}$$

$$\{1, s, r^2, sr^2\} \quad \{1, r, r^2, r^3\}$$

$$2r$$

$$12$$

$$2q$$

$$\{1, (12)(34)\} \triangleleft A_4 \triangleleft S_4$$

$$\begin{array}{l} | \\ \{1, r^2\} \end{array} \quad \begin{array}{l} sr^2s \\ = s s r^{-2} \\ = s^2. \end{array}$$